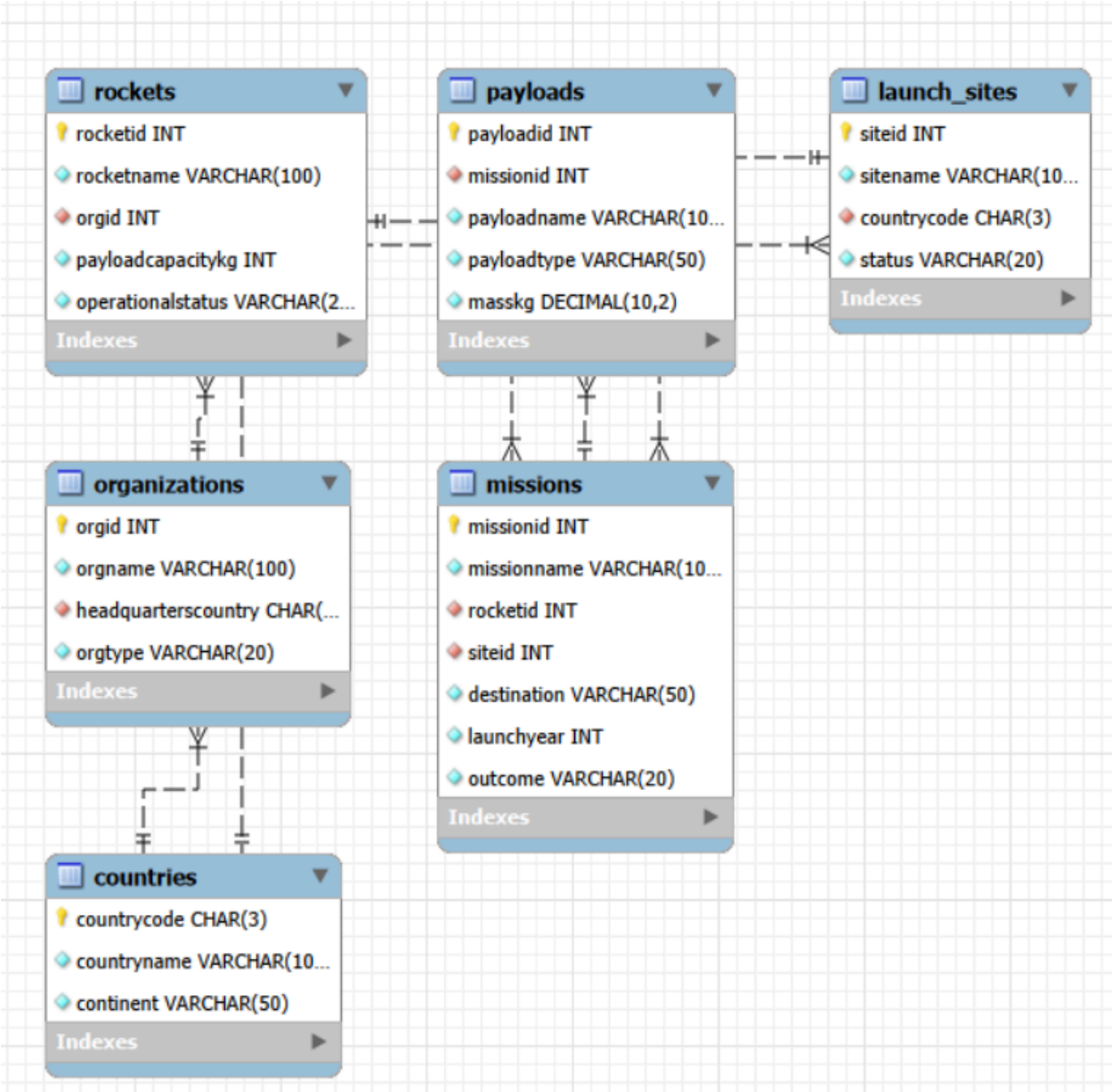
Volodymyr Siedykh
Assignment 4
Generated ER



SQL for each problem

-- 1) Retrieve the mission name, rocket name, organization name, launch site name, destination, launch year, and outcome
-- for all space missions without using table aliases. Sort the results in descending order by launch year and ascending
-- order by mission name.

```
SELECT missions.missionname,
    rockets.rocketname,
    organizations.orgname,
    launch_sites.sitename,
    missions.destination,
    missions.launchyear,
    missions.outcome
FROM missions
INNER JOIN rockets USING(rocketid)
INNER JOIN organizations USING(orgid)
INNER JOIN launch_sites USING(siteid)
ORDER BY missions.launchyear DESC, missions.missionname ASC;
```

missionname	rocketname	orgname	sitename	destination	launchyear	outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

```
-- 2) Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg) for all payloads launched in 2024
-- without using table aliases. Sort the results in descending order by payload mass.

SELECT missions.missionname,
    rockets.rocketname,
    payloads.payloadname,
    payloads.payloadtype,
    payloads.masskg
FROM payloads
INNER JOIN missions USING(missionid)
INNER JOIN rockets USING(rocketid)
WHERE missions.launchyear = 2024
ORDER BY payloads.masskg DESC;
```

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800.00
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500.00
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000.00
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450.00
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200.00
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192.00
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150.00
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100.00

-- 3) Retrieve all launch sites along with their matching mission names, rockets, and launch years for missions that took place in
-- 2024 without using table aliases, ensuring that launch sites with no recorded 2024 missions still appear in the results with
-- null values for the mission fields. Sort the results in alphabetical order by launch site name.

```
SELECT launch_sites.sitename,
      missions.missionname,
      rockets.rocketname,
      missions.launchyear
FROM launch_sites
LEFT JOIN missions ON launch_sites.siteid = missions.siteid AND missions.launchyear = 2024
LEFT JOIN rockets USING(rocketid)
ORDER BY launch_sites.sitename ASC;
```

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

-- 4) Write a SQL query using a Common Table Expression (CTE) to calculate the total payload mass (in kg)
-- deployed by each organization across all missions without using table aliases.
-- Retrieve the organization name, headquarters country, organization type, and total payload mass,
-- and sort the results in descending order by total payload mass.

```
WITH OrgPayloadTotals AS (
   SELECT rockets.orgid, SUM(payloads.masskg) AS TotalPayloadMassKg
   FROM payloads
   INNER JOIN missions USING(missionid)
   INNER JOIN rockets USING(rocketid)
   GROUP BY rockets.orgid
)
SELECT organizations.orgname,
      organizations.headquarterscountry,
```

```
    organizations.orgtype,
    OrgPayloadTotals.TotalPayloadMassKg
FROM organizations
LEFT JOIN OrgPayloadTotals USING(orgid)
ORDER BY TotalPayloadMassKg DESC;
```

orgname	headquarterscountry	orgtype	TotalPayloadMassKg
National Aeronautics and Space Administration	USA	Government	25800.00
Space Exploration Technologies Corp	USA	Private	13692.00
Japan Aerospace Exploration Agency	JPN	Government	6000.00
Roscosmos State Corporation	RUS	Government	2450.00
China National Space Administration	CHN	Government	1200.00
Indian Space Research Organisation	IND	Government	150.00
Rocket Lab	NZL	Private	100.00
Arianespace	FRA	Private	NULL

```
-- 5) Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate total missions by
-- country code and another to calculate total payload mass by country code without using table aliases. Then,
-- combine these CTEs with the countries table to display the country name, total number of missions, and total
-- payload mass (in kg). Sort the results in descending order by total payload mass.

WITH CountryMissionTotals AS (
   SELECT launch_sites.countrycode, COUNT(missions.missionid) AS TotalMissions
   FROM missions
   INNER JOIN launch_sites USING(siteid)
   GROUP BY launch_sites.countrycode
),
CountryPayloadTotals AS (
   SELECT launch_sites.countrycode, SUM(payloads.masskg) AS TotalPayloadMassKg
   FROM payloads
   INNER JOIN missions USING(missionid)
   INNER JOIN launch_sites USING(siteid)
   GROUP BY launch_sites.countrycode
)
SELECT countries.countryname, CountryMissionTotals.TotalMissions,
CountryPayloadTotals.TotalPayloadMassKg
FROM countries
LEFT JOIN CountryMissionTotals USING(countrycode)
LEFT JOIN CountryPayloadTotals USING(countrycode)
```

ORDER BY TotalPayloadMassKg DESC;

countryname	TotalMissions	TotalPayloadMassKg
United States	3	39492.00
Japan	1	6000.00
Russia	1	2450.00
China	1	1200.00
India	1	150.00
New Zealand	1	100.00
France	NULL	NULL
United Kingdom	NULL	NULL